Name _____ Period _____ Date _____

4-4 Lesson Quiz: Adding and Subtracting Rational Expressions

short form: questions 3–5 of the Savvas 4-4 Lesson Quiz. Show your work for every question.

3. Subtract $\dfrac{3}{x-1} - \dfrac{4}{x+3}$.

Ⓐ $\dfrac{-x+13}{(x-1)(x+3)}$; $x \neq -3$ or 1

Ⓑ $\dfrac{-x+5}{(x-1)(x+3)}$; $x \neq -3$ or 1

Ⓒ $\dfrac{7x+5}{(x-1)(x+3)}$; $x \neq -3$ or 1

Ⓓ $\dfrac{7x+13}{(x-1)(x+3)}$; $x \neq -3$ or 1

Work

4. Simplify $\dfrac{\frac{1}{x} - \frac{1}{x+2}}{2}$.

Ⓐ $\dfrac{1}{x+2}$; $x \neq -2$

Ⓑ $\dfrac{1}{x}$; $x \neq 0$

Ⓒ $\dfrac{1}{x(x+2)}$; $x \neq -2$ or 0

Ⓓ $\dfrac{1}{x(x-2)}$; $x \neq 2$ or 0

Work

5. The equation $r = \dfrac{1}{\frac{1}{r_1} + \frac{1}{r_2}}$ represents the total resistance, r, when two resistors whose resistances are r_1 and r_2 are connected in parallel. Find the total resistance when r_1 is x and r_2 is $x + 1$.

Ⓐ $\dfrac{1}{2x+1}$; $x \neq -1$, $-\frac{1}{2}$, or 0

Ⓑ $2x + 1$; $x \neq -1$ or 0

Ⓒ $\dfrac{x(x+1)}{2x+1}$; $x \neq -1$, $-\frac{1}{2}$, or 0

Ⓓ $\dfrac{2x+1}{x(x+1)}$; $x \neq -1$ or 0

Work

Scoring: Questions 3 and 4 are 10 points each; Question 5 is a bonus of up to 5. Total out of 20.

4-4 Lesson Quiz, short form: answer key

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3. **A,** $\frac{-x+13}{(x-1)(x+3)}$**;** $x \neq -3$ **or** 1**.** LCD $(x-1)(x+3)$: $\frac{3(x+3)-4(x-1)}{(x-1)(x+3)} = \frac{3x+9-4x+4}{(x-1)(x+3)} = \frac{-x+13}{(x-1)(x+3)}$. Distractor B $(-x+5)$ writes $-4(x-1)$ as $-4x-4$: the negative was not distributed to the -1. C $(7x+5)$ adds the numerators instead of subtracting: $3(x+3)+4(x-1) = 3x+9+4x-4$. D $(7x+13)$ adds and also writes $4(x-1)$ as $4x+4$: $3x+9+4x+4$. Savvas quiz item 3, DOK 1, skill A86; taught by Example 4. **10 points:** 3 for the LCD, 4 for the numerator with the -1 distributed, 3 for choice A (domain included).

4. **C,** $\frac{1}{x(x+2)}$**;** $x \neq -2$ **or** 0**.** Multiply top and bottom by the LCD $x(x+2)$: $\frac{(x+2)-x}{2x(x+2)} = \frac{2}{2x(x+2)} = \frac{1}{x(x+2)}$. Distractor D has the wrong sign in the factor $x-2$. Savvas quiz item 4, DOK 2, skill A86; taught by Example 6. **10 points:** 3 for the LCD $x(x+2)$, 4 for clearing the small fractions to $\frac{2}{2x(x+2)}$, 3 for choice C.

5. **C,** $\frac{x(x+1)}{2x+1}$**;** $x \neq -1, -\frac{1}{2}$**, or** 0**.** Substitute: $r = \frac{1}{\frac{1}{x}+\frac{1}{x+1}}$. Multiply top and bottom by $x(x+1)$: $\frac{x(x+1)}{(x+1)+x} = \frac{x(x+1)}{2x+1}$. $x = -\frac{1}{2}$ is excluded because it makes $\frac{1}{x}+\frac{1}{x+1} = 0$, so r is undefined there. Distractor D is the reciprocal (the sum $\frac{1}{x}+\frac{1}{x+1}$ before dividing). Savvas quiz item 5, DOK 2, skill A86; Example 6 (the packet's Practice block is its parallel). **Bonus 5:** 2 for the substitution, 3 for choice C with work.

Items left off this form (available for a retake or Form B): Savvas item 1, add $\frac{5}{2x+1} + \frac{8}{4x+2}$, $x \neq -\frac{1}{2}$ (answer A, $\frac{9}{2x+1}$; DOK 1, A86; Example 3 content, practised in Try It 4a and IXL FEX); Savvas item 2, LCM of $(x+3)^2(x-2)$ and $(x+3)(x^2-16)$ (answer D, $(x+3)^2(x-2)(x^2-16)$; DOK 2, A86; Example 2 content, step 1 of Example 4).

IXL replacement: SmartScore 70 or above on FEX and YD8 replaces this quiz with 20/20.